Technical Note: EMR and Wolfram's Computational Universe

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Scope. This note asks whether the effort-motion-result (**EMR**) occurrence at the centre of the State-Dynamic Model of Coexistence (**SDM**) can be represented within Stephen Wolfram's computational-universe approach. It reviews *A New Kind of Science*, the Wolfram Physics Project's hypergraph, multiway, and causal-graph formalisms, and Wolfram's observer-theory proposal. Wolfram's writings are treated as primary sources for that programme, while the Madhyasth Darshan texts remain authoritative for EMR and the SDM remains an analytical reconstruction rather than a quantitative physical theory.

The answer is a qualified **yes**. A complete EMR occurrence can supply the typed semantic content of one local rewrite event, and an ensemble of compatible occurrences can be represented by a causal multiway computation. The synchronization is valid only at the level of **operational semantics**. It does not identify coexistence with computation, *satta* with a hypergraph, a unit with a featureless node, or realised Knowledge with the computational limitations of an observer.

The proposed relation retains the SDM recurrence

$$\mathcal{Q}_n \xrightarrow{\mathrm{Occur}(A_n)} \widetilde{\mathcal{Q}}_{n+1} \xrightarrow{\mathrm{Apply}(\Lambda_n)} \mathcal{Q}_{n+1},$$

while encoding A_n as typed local rewrite events and Λ_n as guarded commit rewrites. Both event sorts contribute to an event-labelled causal and multiway graph.

The Wolfram machinery can therefore execute and compare presentations of an SDM fragment. The SDM must determine the types, guards, identity conditions, and meaning of the events being executed.

1. The computational-universe proposal

1.1 Simple programmes and computational equivalence

A New Kind of Science treats natural and formal processes as computations generated by repeated application of rules. Its Principle of Computational Equivalence proposes that systems whose behaviour is not obviously simple commonly reach an equivalent level of computational sophistication (Wolfram NKS, Basic Framework; Content of the Principle).

Computational irreducibility is the associated claim that the behaviour of many such systems cannot generally be found substantially faster than by following their evolution.

Neither proposition is required in order to encode the SDM. The useful methodological commitment is narrower: local rules may generate global structure, and simulation can expose consequences not evident from inspecting a rule in isolation. The stronger thesis that nearly all nontrivial natural processes possess equivalent computational power remains a Wolfram hypothesis, not a commitment of Madhyasth Darshan or the SDM.

1.2 Hypergraph rewriting

The Wolfram Physics Project represents a state as a relational hypergraph and evolution as local replacement of one relational pattern by another. In its basic presentation, the elements have no intrinsic geometry; spatial and other structure is intended to arise from the relations and their rewriting (Basic Structure). An update event matches a local configuration and replaces it according to a rule. Repeated updates generate successive hypergraphs.

This supplies a useful execution form for the SDM, but only after enrichment. An SDM state contains persistent typed units, kind and order roles, form and internal condition, actual mutualities, containment, body-*jeevan* association, human orientation, and trace records. An untyped hypergraph can encode these by graph gadgets, but a typed attributed hypergraph states the intended distinctions more directly and makes invalid matches easier to reject.

1.3 Multiway systems, causal graphs, and time

When several update events are possible, a multiway system records the alternative event sequences. A causal graph instead records dependencies among events: an event is causally prior to another when the latter depends on structure produced or affected by the former. Wolfram's technical account treats time as the progressive updating of the relational structure and a foliation as a division of the causal graph into successive spacelike slices (Time and Spacetime; Graph Types).

This is close to the SDM's use of an occurrence index. The SDM states that n is a descriptive cut in continuous activity and not a fundamental clock tick. A causal foliation can implement such cuts, provided no metaphysical conclusion is drawn that nature is fundamentally a succession of discrete pauses.

1.4 Causal invariance

A rewrite system is causally invariant when different admissible update orderings yield isomorphic causal graphs. This is stronger than merely reaching an isomorphic final state. The Wolfram Physics Project uses causal invariance to make causal history independent of arbitrary scheduling choices (Causal Invariance; Causal Graphs for Causal-Invariant Rules). An official Wolfram Research bulletin explicitly cautions that confluence and causal invariance are not interchangeable: reconvergence of states does not by itself prove equivalence of causal graphs (Piskunov, "Confluence and Causal Invariance").

That distinction is decisive for synchronization. The SDM cannot assume that compatible local EMR events are schedule-independent. It must make compatibility precise and prove the relevant causal invariance for each implemented fragment.

2. What EMR means in the SDM

2.1 One complete activity, not three clock stages

Madhyasth Darshan describes every physical-chemical activity as the inseparable presence of effort, motion, and result (SB, p. 58). Effort is strength borne in state, motion is operative power, and result is the realised configuration across duration; state and motion are indivisible (SB, pp. 60-62, 248, 256-257; KD 3.8, p. 84; KD 3.11, pp. 102-109). The SDM consequently represents one occurrence as

$$\alpha_{s,n} = \langle \mathrm{Eff}_{s,n}, \mathrm{Mot}_{s,n}, \mathrm{Res}_{s,n} \rangle.$$

The tuple is aspectual, not sequential. A correct computation must not execute effort at one tick, motion at a second, and result at a third. One rewrite event may have a precondition, an update action, and a post-state, but these are parts of its formal presentation; they are not three ontic activities. The EMR record characterises the single complete occurrence through which a persistent unit is present in mutuality.

2.2 Occurrence and structural boundary

The SDM recurrence has two semantic phases:

$$\mathcal{Q}_n \xrightarrow{\mathrm{Occur}(A_n)} \widetilde{\mathcal{Q}}_{n+1} \xrightarrow{\mathrm{Apply}(\Lambda_n)} \mathcal{Q}_{n+1}.$$

A_n contains the compatible complete occurrences of active ontic units and derived association views. Their realised results already form the candidate situation $\widetilde{\mathcal{Q}}_{n+1}$. The second phase examines that candidate for a structural or orientation boundary: closure, disintegration, reproduction, T1 constitutional completeness, body-*jeevan* association or separation, or later human orientation revision. Ordinary displacement, exchange, growth, repair, and immediate consequence are results within Occur; they are not additional structural events.

This distinction should survive implementation. A single undifferentiated rewrite alphabet would make every state change look like the same kind of event. The synchronized system instead needs two typed event sorts, or one compound transaction whose internal phases remain distinguishable:

$$\mathrm{EMREvent} \quad \text{and} \quad \mathrm{BoundaryEvent}\,.$$

The second consumes the candidate results of the first. It does not complete an EMR occurrence that was previously unfinished.

2.3 Persistent bearers and derived scopes

An SDM unit persists through many occurrences. A body-*jeevan* association is a typed relation $j \bowtie (b, r)$, not a merged node, and its public occurrence is derived from the co-present occurrences of body and *jeevan*. A maintained family, work process, or organisation is a relational scope, not a larger sentient or ontic unit. These distinctions constrain the computational representation:

- ontic-unit identity must remain traceable across events;
- a containing activity may be activated at closure without deleting its constituents;
- a body-*jeevan* association must be encoded as an association, not containment or fusion;
- a derived association occurrence must not be counted as a third independent causal event; and
- an organisation may aggregate traces and constraints but must not inherit faculties, *jeevan*, or EMR as if it were a closed unit.

3. The exact correspondence

The correspondence is useful when stated as an interface rather than an identity.

SDM object	Wolfram-style representation	Constraint carried from the SDM
$\mathcal{Q}_n$	Typed attributed relational hypergraph H_n	Nodes and relations retain kind, role, identity, and register
Ontic unit $i \in U_n$	Persistent typed bearer or bearer subgraph	Not reducible to its current adjacency pattern
E_n^U	Typed mutuality hyperedges	Actual facing, not generic connectivity
E_n^R	Typed human-relationship edges between association scopes	Endpoints remain persons-in-association, not organisations
$i \triangleleft_n c$	Immediate containment incidence	Functional, acyclic, and distinct from mutuality
$j \bowtie_n (b, r)$	Typed body-*jeevan* association edge	Neither containment, fusion, nor a new unit
$\alpha_{i,n}$	One typed local update event with an EMR annotation	Effort, motion, and result are co-present aspects

SDM object	Wolfram-style representation	Constraint carried from the SDM
$\alpha_{\eta,n}$	Derived event view over body and *jeevan* events	Adds no independent ontic event
A_n	Compatible event family or causal antichain	Compatibility must be proved, not inferred from locality
$\widetilde{\mathcal{Q}}_{n+1}$	Candidate post-event hypergraph	Already contains realised occurrence results
Λ_n	Guarded structural/orientation rewrite family	Separate event sort with explicit support and writes
L_n	Append-only provenance and consequence subgraph	Later evidence cannot retroactively cause the earlier event
n	Chosen foliation or descriptive slice	No claim of fundamental clock time
$\mathbb{S}$	Standing semantic context outside the rewrite state	Never represented as a node, rule, energy packet, or event

Effort is not best represented as an external input or a quantity injected by the rewrite engine. The typed event annotation describes the qualitative strength borne in the unit's state, read with its order-oriented *svabhav-dharma* aspect; it is not an additional guard. Motion is the operative relational and configurational change realised by the event. Result is the committed result-side projection that becomes present in the candidate state. These three fields must be jointly validated against one occurrence; none should be scheduled independently.

4. A synchronized event semantics

4.1 Encoding a bounded situation

Let $H(\mathcal{Q})$ encode the bounded situation while preserving its registers. A rewrite rule has a typed left pattern L_e, a typed right pattern R_e, and a match

$$m_e : L_e \longrightarrow H(\mathcal{Q}_n).$$

The match is admissible only when it preserves kind, base role, identity, actual mutuality, containment, association, and any relevant conformance conditions. The rule then yields an event record

$$e_i = \langle i, m_e, \mathrm{Eff}_{i,n}, \mathrm{Mot}_{i,n}, \mathrm{Res}_{i,n}, \mathrm{read}(e_i), \mathrm{write}(e_i)\rangle.$$

The read and write sets support dependency analysis. They are implementation metadata, not additions to the Madhyasth Darshan ontology. The event is well formed only when its

EMR projection is one admissible $\alpha_{i,n}$ of the source unit.

Where no represented coordinate changes, the engine still needs a labelled state-preserving occurrence in the causal trace. Omitting it would identify absence of encoded change with absence of unit-activity. Such identity events can be hidden from a selected display, but not from a claim to represent every active unit's complete occurrence.

For an association scope $\eta = (j, b, r)$, the implementation derives e_η from e_j, e_b, the two interface channels, and the active association. It must not schedule e_η as an event independent of both bearers. This avoids double-counting a public joint occurrence in the causal graph.

4.2 Constructing the causal graph

Define a causal dependency

$$e \prec e' \quad \text{when} \quad \mathrm{write}(e) \cap \big(\mathrm{read}(e') \cup \mathrm{write}(e')\big) \neq \varnothing,$$

subject to an explicit commutation rule for mergeable writes. The transitive reduction of this dependency relation gives a causal graph for the implemented trace. A compatible occurrence ensemble corresponds to an antichain, or to a set whose internal overlaps have a defined commutative merge.

Structural events are causally downstream of the EMR events whose candidate results satisfy their guards. A closure event, for example, depends on the constituent occurrences that make $\mathrm{CandidateClosed}$ true. Its effect activates a containing activity and its immediate-containment incidences while preserving constituent identity. It does not replace those occurrences with a single untyped node.

4.3 Defining synchronization

Let $\approx_{\mathrm{SDM}}$ mean observational equivalence with respect to the declared SDM fragment. At minimum it preserves active unit identities, kinds and roles, form and internal condition, actual mutualities, immediate containment, body-*jeevan* associations, human orientation where in scope, and retained trace records.

For two valid schedules σ and τ of the same compatible event family, synchronization requires

$$\mathrm{Exec}_\sigma(H(\mathcal{Q}_n)) \approx_{\mathrm{SDM}} \mathrm{Exec}_\tau(H(\mathcal{Q}_n)),$$

and, for causal invariance, isomorphism of their event-labelled causal graphs. Equality of final graph shape alone is insufficient. Labels may be quotientable when they are merely presentation artifacts, but unit identity, event sort, source type, and provenance cannot be discarded by the isomorphism.

The synchronized computation is therefore not simply "run all matching rules." It is "run all **typed, semantically admissible** events, preserve the SDM registers, and compare schedules under an SDM-defined equivalence."

5. Causal invariance and natural orderliness

5.1 Where causal invariance is a plausible requirement

Material and pranic fragments governed by constitution- or seed-conformance are the strongest candidates for causal invariance. If two local occurrences have disjoint support, or if their overlapping writes have a stated commutative merge, their order should not change the causal account of the bounded result. Proving that property would give computational content to the SDM's compatibility condition without claiming that causal invariance is the meaning of *niyam* or *niyati-kram*.

The correspondence is limited:

- *niyam* names definite lawfulness in conduct; causal invariance is a property of a rule system under rescheduling.
- *niyati-kram* restricts admissible structural reachability; it is not the multiway graph itself.
- *dharma* states order-relative invariant orientation and fulfilment; it is not a rewrite guard, conserved bit, or objective function.
- complementarity is a typed semantic judgement evidenced in offering, acceptance, and fulfilment; it is not mere pattern overlap.

Causal invariance may demonstrate that an implementation does not depend on arbitrary update order. It does not prove the darshan's claim that nature is lawful, regulated, and balanced.

5.2 Critical pairs are diagnostic

Overlapping local rules generate critical pairs. Each pair should be classified as:

1. independent and causally invariant;
2. mergeable under an explicit typed rule;
3. an authentic alternative outcome admitted by the fragment;
4. an invalid branch created by a missing type or conformance guard; or
5. unresolved because the SDM and its sources do not determine the case.

This audit is more useful than assuming confluence. A non-reconvergent critical pair may reveal an implementation error, but it may also expose a real distinction such as closure versus mixture, association versus separation, or correct versus failed human fulfilment.

6. Multiway branching and human fallibility

The multiway graph has two different roles and they should not be conflated. First, it represents alternative schedules of events that are intended to describe the same bounded occurrence. Causal invariance is appropriate here. Second, it represents substantively different admissible selections or outcomes. These branches need not reconverge.

The second role becomes essential in the human order. Actual relationship, accepted relationship-model, operative recognition, selection, conduct, consequence, and later evaluation can diverge. Forcing every human branch to reconverge would erase the very possibility of error, injustice, learning, or refinement that the SDM is designed to preserve. A human multiway model should therefore distinguish:

- **schedule branches**, which ideally quotient under causal invariance;
- **choice branches**, which record different selections under the same situation;
- **epistemic branches**, which arise from different operative recognitions or unavailable evidence; and
- **consequence branches**, which remain distinct when conduct produces different bodily, relational, or natural results.

The existence of a computational branch does not make that branch ontically realised. A multiway graph is a model of possibilities under stated rules. The ruliad - Wolfram's proposed limit of all possible computational rules and applications - is unnecessary for the synchronization and should not be imported as an ontology of coexistence. SDM types, conformance, complementarity, structural guards, and retained evidence select a narrow model family rather than treating all formal rewrites as equally real.

7. Observer theory and SDM trace semantics

Wolfram's observer theory treats observation as computationally bounded equivalencing: an observer groups many underlying states into the same perceived state while computation continues to generate distinctions (Wolfram, "Observer Theory"). This offers a useful model for three downstream SDM tasks:

- defining which coordinates of $\mathcal{Q}_n$ are available in an access domain;
- specifying coarse-grained observation ledgers over causal traces; and
- comparing whether two computational presentations are indistinguishable to a declared observer interface.

It does not model realised Knowledge by itself. A bounded computational observer is characterised by what it cannot distinguish. The SDM's human account distinguishes actual relationship from accepted model, operative recognition, knowing, believing, fulfilling, consequence, later evaluation, and realisation. Coarse-graining can represent limited access to evidence, but it cannot establish *jeevan*, enlightenment, truth-realisation, or mutual satisfaction. Observer equivalence belongs in the trace and verification layer, not in the ontological definition of the human being.

Computational irreducibility has a similarly modest role. It gives a reason to simulate long event traces rather than expect a closed-form shortcut. It does not supply missing transition laws, numerical parameters, measurements, or empirical validation. The SDM remains qualitative until such specifications are independently justified.

8. What must not be synchronized

Several non-identities are constitutive of the proposal.

Wolfram-oriented identification	SDM decision	Reason
Existence is computation	Reject	The SDM begins from the coexistence of actionless *satta* and state-dynamic units
Satta is the hypergraph or rewrite substrate	Reject	*Satta* is not a member of state, an event, a rule, or a transmitted resource
A unit is a featureless node	Reject	A unit is a persistent bounded activity with kind, condition, and order-specific aspects
EMR is match, rewrite, then output as three stages	Reject	Effort, motion, and result are co-present aspects of one occurrence
Every relation is adjacency	Reject	Mutuality, containment, association, and human relationship have different types
Closure is node fusion	Reject	A containing activity is established while constituent identities persist
Body-*jeevan* is graph contraction	Reject	Association manifests a joint role without producing a third ontic unit
An organisation is a super-node with faculties	Reject	Maintained order is relational and enacted by participants
Causal invariance is *niyam*	Reject	One is a scheduling property; the other is a claim of definite lawfulness
Computational equivalence proves sameness of orders	Reject	The four orders retain different conformance and fulfilment conditions
Observer coarse-graining is realised Knowledge	Reject	Limited equivalencing does not establish knowing, fulfilment, or realisation
Every multiway branch exists	Reject	Branches are computational possibilities until an interpretation is justified

These refusals do not weaken the computational programme. They define the interface that keeps an implementation from silently replacing the theory it is meant to explore.

9. First implementation: material structural closure

The first pilot should use the material structural-closure fragment. It is small enough to audit, avoids premature claims about the body-*jeevan* interface, and aligns with the first pilot recommended in the companion MD-TOPOS note.

The host state should include persistent unit identity, material kind, form and internal condition, actual typed mutuality, immediate containment, and active containing activities. The first event alphabet should contain:

- typed material EMR occurrences whose results update candidate conditions and relations;
- `ConstructClosure`, guarded by `CandidateClosed` and a whole-part consistency constructor; and
- `Disintegrate`, guarded by failure of continued candidate closure.

The pilot should produce positive terminal tags such as `Closed`, `Nonclosing`, and `Disintegrated`, rather than infer every negative claim from absence. It should then perform four tests.

1. **Type preservation:** no rule confuses a constituent, containing activity, mutuality edge, or containment edge.
2. **EMR atomicity:** the engine records one event with three co-present projections, not three scheduled events.
3. **Schedule comparison:** two compatible update schedules reach SDM-equivalent candidate and committed states and have isomorphic labelled causal graphs.
4. **Discriminating failure:** at least one overlapping rule pair fails causal invariance until a missing guard or merge rule is supplied.

The pilot succeeds if the formalism detects when two implementations preserve the same typed occurrence semantics and when a superficially similar rewrite loses an SDM distinction. It does not need to reproduce physical spacetime, derive numerical laws, or claim that the implemented rule is nature's fundamental rule.

10. Synchronization decision

Wolfram's computational-universe approach and the SDM can be synchronized as follows:

Layer	Decision
Local rule execution	Adopt through typed attributed rewriting
EMR	Encode as the three co-present semantic projections of one event

Layer	Decision
Concurrent A_n	Represent as a compatible event family and causal antichain
Occurrence index	Interpret as a chosen causal foliation or descriptive cut
Structural Λ_n	Encode as a second guarded event sort downstream of candidate results
Multiway graph	Use for schedule alternatives and genuine outcome branches, explicitly distinguished
Causal invariance	Require and prove for fragments intended to be schedule-independent
Computational irreducibility	Use as motivation for simulation, not as evidence for the ontology
Observer theory	Use for typed access and coarse-grained trace interfaces
Principle of Computational Equivalence	Leave optional and external to the SDM
Ruliad metaphysics	Do not import

The resulting system is best described as a **typed causal multiway implementation of SDM occurrence semantics**. Wolfram supplies an event calculus, causal ordering, branching representation, and powerful questions about schedule independence. EMR supplies the content of an event; the SDM supplies the bearer types, standing context, admissibility conditions, structural distinctions, human fallibility, and standards of trace equivalence. The synchronization is productive precisely because it remains asymmetric: computation presents and tests the SDM fragment, while the SDM determines what the computation means.

Editorial Notes

Relation to the retired first-principles note

An earlier internal note compared Wolfram's programme with *Coexistence From First Principles*. The present note does not carry forward that retired kernel or its symbols. It reconstructs the comparison from the current SDM recurrence, complete occurrence $\alpha_{s,n}$, bounded situation $\mathcal{Q}_n$, candidate result, structural event set Λ_n, containment $\triangleleft$, body-*jeevan* association $\bowtie$, and later trace log L_n. The continuing conclusion is limited to the use of Wolfram-style computation as an implementation and exploration layer.

Status of the formal correspondence

The event encoding, causal dependency relation, SDM observational equivalence, and proposed pilot are analytical constructions of this note. They are not formulations found in the Madhyasth Darshan texts or the Wolfram sources. The primary texts support the inseparability and aspectual alignment of effort, motion, and result; the SDM supplies the typed recurrence; Wolfram's sources supply the rewrite, multiway, causal-graph, causal-invariance, irreducibility, and observer concepts. A working implementation and causal-invariance proof remain future work.

Causal invariance is not confluence

This note follows the narrower Wolfram Research definition: causal invariance requires isomorphic causal graphs across admissible event orderings. It does not infer that property from reconvergence of final states. The proposed pilot therefore checks both SDM-level state equivalence and event-labelled causal-graph equivalence.

References

State-dynamic reconstruction

- **SDM** - *From Unit Activity to Human Orderliness: A State-Dynamic Reconstruction of Coexistence*. Used: EMR and continuing transition (§§1-2); order-*dharma*, structural closure, body-*jeevan* association, maintained order, and human fallibility (§§3-8); model limits (§12); bounded situation, complete occurrence, recurrence, structural events, and later evaluation (Appendix A).
- **MD-TOPOS note** - *Technical Note: MD-TOPOS and the State-Dynamic Model of Coexistence*. Used: typed implementation boundary, scoped observation ledgers, and recommendation of the material structural-closure pilot (§§5-10).

Primary Madhyasth Darshan texts

- **MVD** - Nagraj, A. *Madhyasth Darshan - Co-existentialism*. English translation by Rakesh Gupta. Cited: time as duration of activity and the indivisibility of state and motion (pp. 34, 40; §1.3, §2); unit, mutuality, form, property, essential nature, and *dharma* (pp. 42, 47-50; §§2-3, §8); actionless Omnipresence and saturated units (pp. 35-36, 48, 174; §8).
- **SB** - Nagraj, A. *Samadhanatmak Bhautikvad (Resolution Centred Materialism)*. English translation by Rakesh Gupta. Cited: inseparable effort-motion-result and its alignment with strength, power, form, property, essential nature, and *dharma* (pp. 58, 60-62, 248, 256-257; §2-§4); order-specific recognition, complementarity, and balance (pp. 143-144, 231, 235, 249-251, 255; §5).

- **JV** - Nagraj, A. *Jeevan Vidya: An Introduction*. English translation by Rakesh Gupta. Cited: body-*jeevan* association, four orders, recognition, fulfilment, human relationship, evaluation, and mutual satisfaction (pp. 47-49, 59, 69-70, 97-98, 137-139; §§2-3, §6-§8).
- **KD** - Nagraj, A. *Manav Karm Darshan*. Working English translation; Hindi source `KD-karm darshan v5.pdf`. Machine-assisted working translation, cited as corroboration only: state-motion indivisibility (3.8, p. 84) and strength, power, result, pressure, and flow (3.11, pp. 102-109; §2).

Stephen Wolfram and computational universe

- **Wolfram NKS** - Wolfram, Stephen. *A New Kind of Science*. Wolfram Media, 2002. Online edition: Basic Framework and Content of the Principle of Computational Equivalence. Cited: processes as computation, computational equivalence, and computational irreducibility (§1.1, §7).
- **Wolfram Physics: basic structure** - Wolfram Physics Project. *A Class of Models with the Potential to Represent Fundamental Physics*, Basic Structure. Cited: relational hypergraphs and local rewriting (§1.2).
- **Wolfram Physics: time and graphs** - Wolfram Physics Project. Time and Spacetime and Appendix: Graph Types. Cited: causal graphs, multiway graphs, time through updating, and foliations (§1.3, §3-§4).
- **Wolfram Physics: causal invariance** - Wolfram Physics Project. Causal Invariance and Causal Graphs for Causal-Invariant Rules. Cited: schedule-independent causal graphs (§1.4, §4-§6).
- **Piskunov 2020** - Piskunov, Max. "Confluence and Causal Invariance". Wolfram Physics Project Bulletin, November 16, 2020. Cited: distinction between state confluence and causal invariance (§1.4, §4-§5, Editorial Notes).
- **Wolfram Observer Theory** - Wolfram, Stephen. "Observer Theory". *Stephen Wolfram Writings*, December 29, 2023. Cited: computationally bounded observation and equivalencing (§7).